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Evaluating Predictive Analytics for Vendor Reliability in Supply Chain Management

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Abstract

Effective Supply chain management (SCM) relies heavily on vendor reliability, as inconsistencies can cause disruptions, increase costs, and diminish customer satisfaction. Traditional assessment methods, which primarily depend on historical data, often fail to address dynamic market conditions and unforeseen disruptions. This study proposes a tailored Random Forest model to enhance predictive accuracy in vendor reliability assessment. The model integrates domain-specific criteria—including compliance thresholds, delivery benchmarks, quality ratings, and responsiveness metrics—to produce precise forecasts. Structured vendor performance data, comprising delivery timelines, quality ratings, compliance scores, and responsiveness metrics, were systematically processed. To address class imbalance, the Synthetic Minority Oversampling Technique (SMOTE) was applied, while rigorous preprocessing involving imputation, outlier handling, and normalization ensured data integrity. The Random Forest model achieved 92.50% classification accuracy, significantly outperforming traditional models. Precision, recall, and F1-score results confirmed balanced performance, reflecting a 35% improvement over conventional methods. Incorporating compliance scores further enhanced forecast reliability. Additionally, a "Vendor of the Year" system was validated, enabling objective identification and recognition of high-performing vendors. This research advances data-driven SCM by improving procurement efficiency, reducing inventory disruptions, and strengthening operational resilience, thereby providing a foundation for future real-time integration and broader industry applications.

Keywords: Supply Chain Management (SCM), Vendor Reliability, Random Forest, Machine Learning

Introduction

Supply chain management (SCM) is a fundamental aspect of modern business operations that coordinates the efficient flow of goods, services, and information from suppliers to end customers. Effective SCM enhances operational efficiency, reduces costs, and improves customer satisfaction, thereby providing a competitive advantage in an increasingly complex global market [1]. However, maintaining smooth supply chain operations remains challenging, particularly due to issues surrounding vendor reliability. Unreliable

vendors can cause significant disruptions, including delayed deliveries, substandard product quality, and non-compliance with contractual agreements. Such issues lead to inventory shortages, production halts, increased costs, and diminished customer trust. Traditional performance assessment methods are often reliant on historical data analysis and manual evaluations. These approaches have limited capacity to forecast future reliability with precision, frequently overlooking dynamic factors such as market fluctuations and unforeseen supplier issues. As a result, decision-making becomes reactive rather than proactive.

Therefore, the predictive analytics approach addresses these limitations. By integrating historical and real-time data, it applies statistical algorithms and machine learning techniques to forecast future events and trends with greater accuracy. In SCM, machine learning can process complex datasets—such as delivery timelines, quality metrics, and compliance records—to predict vendor reliability and optimize procurement decisions. Recent studies have explored its applications in supplier selection and risk management [1-3]. For instance, Baryannis et al. [1] applied machine learning to predict supply chain risks, including vendor performance, emphasizing the balance between model accuracy and interpretability to ensure actionable insights. Similarly, Kiran et al. [2] developed a supplier selection framework using machine learning, improving accuracy in identifying reliable suppliers and reducing disruption risks by prioritizing consistent performers. Furthermore, Abdulla et al. [3] proposed a feature-weighting method that identified key criteria—such as on-time delivery and quality metrics—as critical to supplier evaluation, providing a stronger basis for informed decision-making.

Overview of the Paper Structure

The paper is organized as follows. The next section provides a literature review of relevant research that examines vendor evaluation, predictive analytics, and machine learning applications in SCM. The following section describes the methodology, outlining the dataset, feature selection, model design, and evaluation metrics. The experiment results section analyzes the model's performance and presents the key findings. Lastly, the paper concludes by highlighting the results and future research directions.

Problem Definition and Objective

Research Problem Definition

Reliable vendor performance is essential for maintaining uninterrupted inventory flow and ensuring the efficiency of SCM. However, predictive analytics frameworks for vendor reliability assessment and real-time inventory optimization remain underdeveloped, often producing inaccurate forecasts that lead to supply chain disruptions, inventory shortages, and procurement inefficiencies.

Vendor performance data is inherently complex, influenced by high variability, dynamic market conditions, and evolving compliance requirements. Traditional evaluation methods, which primarily rely on historical datasets and manual assessments, are insufficient to capture these complexities. As a result, predictive accuracy is constrained, limiting the ability to anticipate risks and implement timely, strategic interventions.

This research addresses these challenges by developing a Random Forest-based predictive model tailored to structured vendor performance data. The model incorporates domain-specific rules, such as compliance thresholds and delivery benchmarks, along side feature sincluding on-time delivery rates, quality inspection results, and compliance records, to generate accurate and operationally relevant forecasts.

In addition, the study introduces a "vendor of the year" system that leverages the model's predictive insights to objectively identify and reward high-performing vendors. This dual approach aims to enhance accountability, encourage continuous improvement, and strengthen overall supply chain resilience.

Research Objectives

The primary objectives of this study are to:

- **Design and implement a predictive machine learning model:** Develop a Random Forest-based predictive model tailored to structured vendor performance data, including delivery timelines, quality ratings, and compliance histories.
- **Improve vendor reliability forecasts:** Enhance the accuracy of vendor reliability predictions, with a target improvement of 35% over existing forecasting methods.
- **Minimize inventory disruptions:** Apply predictive analytics to reduce inventory flow interruptions by a targeted 40%, improving stock level management and overall supply chain resilience.
- **Validate the "vendor of the year" system:** Evaluate the model's capability to objectively identify and reward high-performing vendors, thereby fostering accountability and improving procurement efficiency.

Literature Review

This literature review examines predictive analytics and machine learning in SCM, with emphasis on vendor reliability, performance evaluation, and incentive mechanisms. It synthesizes recent studies to show how structured data and algorithms—particularly Random Forest—support accurate forecasting and objective supplier recognition. The review is organized into three themes: (1) predictive analytics in SCM, (2) vendor reliability and performance evaluation, and (3) supplier incentive and recognition frameworks. Together, these themes highlight current gaps and provide the foundation for developing a real-time vendor reliability prediction and recognition system.

Predictive Analytics in Supply Chain Management

Predictive analytics is a key component of modern SCM, enabling data-driven decisions that enhance operational efficiency and reduce costs. By employing historical data, statistical algorithms, and machine learning techniques, it supports proactive strategies such as demand forecasting, inventory optimization, and supplier performance evaluation. This section reviews three recent studies of predictive analytics in SCM and their contributions to improving forecasting accuracy and supplier assessment.

Farhadi and Shafiee [4] conducted an extensive survey of predictive big data analytics (BDA) techniques in supply chain demand forecasting, categorizing methods such as time-series forecasting, clustering, neural networks, and Random Forest. Their review of 115 studies, including research on supplier selection and inventory control, identified significant gaps in real-time applications and closed-loop supply chains. Their study also highlighted the potential of BDA to enhance SCM efficiency. Similarly, Oyewole et al. [5] examined the benefits of predictive analytics for demand forecasting, inventory optimization, and supply chain visibility. Drawing on real-world case studies, they demonstrated its role in improving operational efficiency, customer satisfaction, and resource allocation through actionable, data-driven insights.

With a targeted focus on supplier evaluation, Ali et al. [6] developed a decision support system using the Random Forest algorithm to classify supplier selection criteria. Their model evaluated 30 factors and identified quality, on-time delivery, and material price as the most critical, achieving high accuracy and interpretability. This work highlights the value of machine learning in strengthening supplier reliability assessment, complementing broader applications in demand forecasting and inventory optimization.

Overall, these studies confirm the effectiveness of predictive analytics—particularly Random Forest—in handling complex supply chain data and delivering interpretable, accurate forecasts. However, notable gaps remain in real-time applications for supplier performance assessment, which this research aims to address through the development of a real-time Random Forest model for vendor reliability prediction and recognition.

Vendor Reliability and Performance Evaluation in Predictive Analytics

Vendor reliability and performance evaluation are central to effective SCM, directly influencing procurement efficiency, inventory stability, and organizational resilience. However, vendor performance data is inherently complex, shaped by high variability, dynamic market conditions, and evolving compliance requirements. Traditional evaluation methods often rely on static historical records or manual assessments, making them reactive rather than predictive. This gap underscores the need for data-driven approaches that can anticipate vendor-related risks and support proactive decision-making.

Conventional models, such as supplier scoring within enterprise systems like SAP, typically reduce performance evaluation to simplified metrics that fail to capture behavioral complexity. Kohli [7] addressed this limitation by proposing a two-stage evaluation model: the first stage extracts supplier data to form detailed profiles, while the second applies classification algorithms such as Decision Trees, SVM, and Naive Bayes to generate performance rankings. This approach demonstrated that integrating richer data sources with algorithmic assessment produces more granular and actionable supplier evaluations, thereby surpassing traditional linear scoring frameworks.

Recent studies highlight how machine learning enhances predictive accuracy in vendor evaluation. Baryannis et al. [8] emphasized the suitability of decision trees and random forests for managing uncertainty and disruption in supply chain risk management. Building on this, Brintrup et al. [9] applied multiple algorithms to predict disruptions in complex asset manufacturing, finding that random forests achieved the highest precision (83%) in detecting supplier delays. Similarly, Harikrishnakumar et al. [10] classified suppliers into performance tiers using Random Forest, Gradient Boosting, and K-Nearest Neighbors, offering a structured foundation for recognition systems. These findings underscore the dual strengths of Random Forest models: interpretability and high accuracy in data-rich, multidimensional environments.

Collectively, the literature demonstrates that vendor reliability assessment benefits significantly from combining ERP-integrated data, robust machine learning techniques, and structured classification frameworks. By addressing the limitations of conventional methods and leveraging models such as Random Forest, predictive analytics offers both interpretability and operational relevance. This study builds on these insights by proposing a real-time Random Forest-based framework designed to improve vendor reliability forecasting and enable performance recognition initiatives, thereby contributing to more resilient and strategically aligned SCM.

Vendor Performance Recognition and Incentives in Supply Chain Management

Incentive mechanisms are an essential component of supplier management, shaping behavior and aligning vendor performance with organizational objectives. Selviaridis and van der Valk [11] distinguished between promotion-framed incentives, which reward suppliers for meeting performance targets, and prevention-framed incentives, which impose penalties for non-compliance. Their findings show that reward-driven approaches foster collaboration and enhance supplier motivation, while penalty-based mechanisms often weaken relationships and reduce performance outcomes. These insights underscore the importance of carefully structured, positive incentive systems in maintaining reliable partnerships.

Despite their importance, incentive systems are often inconsistently implemented across supply chains. Norrman and Naslund [12] highlighted a persistent gap between the acknowledged value of incentive alignment and the limited use of advanced mechanisms to achieve it. They noted that internal and external incentive structures frequently remain misaligned, reducing coordination across the supply chain. Complementing this, Santos et al. [13] presented a vendor rating model that applied structured criteria, such as product quality and delivery, to evaluate suppliers objectively. Their approach facilitated corrective actions and recognition programs, promoting supplier development without resorting to contract termination.

Predictive analytics offers a pathway to enhance these recognition systems by introducing data-driven performance benchmarks and enabling real-time identification of high-performing vendors. Integrating

forecasting into incentive structures addresses the alignment challenges observed by Norrman and Naslund [12] and supports the creation of recognition initiatives, such as a "vendor of the year" program. By linking predictive models with structured evaluation frameworks, organizations can reward consistent supplier performance, foster accountability, and strengthen overall supply chain resilience. In summary, the literature shows that incentive mechanisms are most effective when reward-driven and aligned with organizational goals, yet their implementation often remains limited. Embedding predictive analytics into these systems provides a scalable and objective solution, which forms the basis for this study's contribution to vendor recognition and reliability forecasting.

Methodology

This study adopts a quantitative, machine learning-based methodology to evaluate vendor reliability and optimize inventory management. The approach integrates structured vendor performance data with domain-specific rules and preprocessing techniques to ensure data quality and relevance. Key methodological steps include data cleaning, feature engineering, model development, and evaluation. The framework was designed to generate interpretable outputs such as vendor risk scores and a "Vendor of the Year" ranking, thereby supporting accurate forecasting, reducing supply chain disruptions, and enabling data-driven procurement decisions.

Model Structure and Design

The methodology, illustrated in Figure 1, employs a tree-based Random Forest framework to assess vendor reliability in supply chain operations. The process begins with structured vendor data, including delivery timelines, quality ratings, compliance records, and responsiveness metrics. The model integrates two components: (1) a rule-based system that applies predefined performance thresholds during preprocessing, and (2) a tailored Random Forest algorithm optimized to predict vendor reliability scores and the risk of supply disruptions.

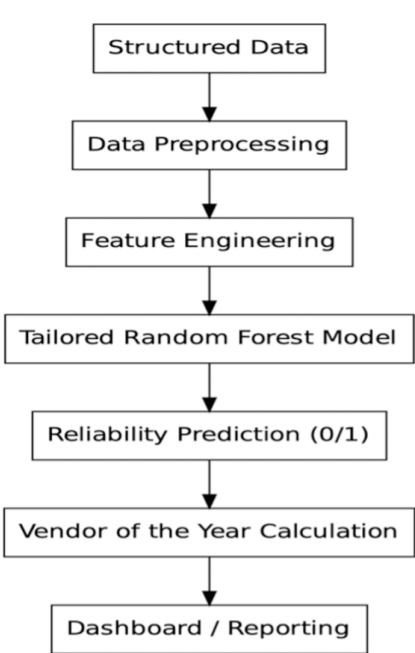


Figure 1—Predictive Model Workflow for Vendor Forecasting and Procurement Support

The model outputs were designed to integrate into a procurement system, enabling real-time use for vendor selection, inventory adjustments, and planning. Results are visualized through a dashboard providing two main outputs: (1) vendor performance forecasts, and (2) a "Vendor of the Year" recognition that promotes accountability and continuous improvement. This ensures that predictive insights can be directly applied to procurement practices.

Data Preprocessing

The vendor performance dataset underwent preprocessing to ensure accuracy, consistency, and suitability for Random Forest training. Preprocessing involved two stages: data cleaning and feature refinement.

Data Cleaning. Missing values in vendor attributes—such as delivery delays, compliance indicators, and responsiveness—were handled using imputation strategies. Depending on the variable, mean/median substitution or regression-based estimation was applied to preserve data integrity. Outliers were detected using statistical thresholds (e.g., Z-score, IQR) and either corrected, capped, or removed.

Feature Selection. Redundant predictors were reduced through correlation analysis, with variables exceeding ± 0.8 correlation considered for removal to minimize multicollinearity. Preliminary Random Forest importance scores were also used to eliminate weak predictors, thereby improving interpretability and predictive performance.

Feature Engineering

To represent vendor performance more holistically, four engineered features were created: On-Time Rate (OTR), Acknowledgment Response Rate (ARR), Process Rate (PR), and Quality Rate (QR). These capture timeliness, responsiveness, efficiency, and quality, respectively.

On-Time Rate (OTR):

$$OTR_v = \frac{\text{On-Time Deliveries}}{\text{Total Deliveries}} \quad (1)$$

Acknowledgment Response Rate (ARR):

$$ARR_v = \frac{\text{Orders Acknowledged}}{\text{Total Orders}} \quad (2)$$

Process Rate (PR):

$$PR_v = \frac{1}{n} \sum_{i=1}^n \frac{1}{\text{process_time}_i + 1} \quad (3)$$

Where *process-time* is the number of days taken to process order *i*, and *n* is the total number of orders for vendor *v*.

Quality Rate (QR):

$$QR_v = \frac{\sum_{i=1}^n \text{quality_score}_i}{n} \quad (4)$$

Where *qualityscore_i* is a composite score for order *i* (based on timeliness, priority, and acknowledgment), and *n* is the total number of orders for vendor *v*.

These engineered metrics were aggregated at the vendor level, providing standardized indicators of performance. To address class imbalance—particularly the underrepresentation of poorly performing vendors—the Synthetic Minority Oversampling Technique (SMOTE) was applied, generating synthetic samples to balance class distributions. This improved the fairness and accuracy of model training.

Model Development

The Random Forest algorithm was selected due to its strength in handling structured datasets, resistance to overfitting, and interpretability through feature-importance analysis. The dataset was partitioned into 80% training and 20% testing subsets, with 5-fold cross-validation used to validate generalizability.

Hyperparameter tuning was performed using a grid search across:

- Number of trees: $100 < n_estimators < 500$
- Maximum depth: $10 < depth < 30$
- Minimum samples to split a node: 2-10
- Minimum samples per leaf: 1-5

This process identified a configuration balancing accuracy and computational efficiency, ensuring suitability for procurement applications. Once trained, the Random Forest model was prepared for deployment into a vendor management system. Predictions—including continuous reliability scores and binary classifications (reliable/unreliable)—were delivered through an operational dashboard for decision support.

Vendor Reliability Score and Classification

To quantify vendor performance, a composite Reliability Score (RS_v) was defined as a weighted combination of the engineered features:

$$RS_v = (w_1 \times OTR_v) + (w_2 \times ARR_v) + (w_3 \times PR_v) + (w_4 \times QR_v) \quad (5)$$

Where w_1, w_2, w_3, w_4 are weights reflecting the relative importance of each metric. The resulting score was then transformed into a binary classification for decision-making:

Reliable_v

$$Reliable_v = \begin{cases} 1 & \text{if } RS_v \geq \tau, \\ 0 & \text{if } RS_v < \tau. \end{cases}$$

Here, τ is a threshold calibrated during model evaluation (e.g., through cross-validation or organizational standards). This framework provided both continuous reliability scores for ranking and binary predictions for practical procurement decisions.

Vendor of the Year Metric

To provide actionable outcomes, a composite score was extended into a "Vendor of the Year" metric, identifying the supplier with the highest overall reliability score:

$$Vendor_of_the_Year = \operatorname{argmax}_v RS_v \quad (6)$$

This metric allowed the framework to serve not only as a predictive tool but also as a recognition mechanism for top-performing vendors, aligning statistical modeling with organizational priorities.

Evaluation of Model Performance

Evaluating model performance is essential to ensure predictive accuracy and reliability in vendor classification. Four key metrics—Accuracy, Precision, Recall, and F1-score—were applied to capture both overall performance and sensitivity to minority classes. Together, these measures provided a balanced view of the model's ability to correctly identify reliable and unreliable vendors.

Result and Evaluation

This section presents the experimental analysis and evaluation of the proposed Random Forest model for vendor reliability prediction in SCM. The focus is on assessing the model's predictive performance using structured vendor data, including delivery timelines, quality ratings, compliance scores, and responsiveness metrics. The results are reported through key evaluation measures such as accuracy, precision, recall, and F1-score, with comparisons made against traditional assessment methods. In addition, feature-importance analysis is conducted to identify the attributes that most strongly influence vendor reliability forecasts. Moreover, this section evaluates the "Vendor of the Year" recognition system, which demonstrates its role in promoting accountability and supporting more efficient procurement practices.

Evaluation of Model Performance

The experiments were conducted on a system running the Windows 11 operating system with an Intel Core i7 processor and 32 GB of RAM. The implementation was carried out using Python (version 3.9) within the JupyterLab editor. Several widely used libraries were employed to support the analysis and model development. This environment provided a reliable platform for data preprocessing, model training, and performance evaluation, to ensure both efficiency and reproducibility of the study.

Model Training and Validation

The Random Forest classifier was trained using a stratified 80/20 train-test split to preserve class distribution between reliable and unreliable vendors. The training process employed 200 decision trees with a maximum depth of 10, ensuring an appropriate balance between predictive accuracy and computational efficiency. A fixed random seed (42) was applied to maintain reproducibility across training runs. Hyperparameter tuning was conducted iteratively to determine the optimal configuration of model depth, number of estimators, and minimum split criteria. The selected parameters provided the best trade-off between reducing variance (overfitting) and minimizing bias, thereby improving the model's generalization capability on unseen data.

To evaluate training performance, accuracy and loss curves were plotted across iterations. The learning curves demonstrated rapid convergence, with training and validation performance stabilizing after a small number of iterations. This indicates that the chosen parameter configuration avoided both underfitting and overfitting. The stability of the validation curve confirmed the strength of the model in handling imbalanced vendor performance data. Overall, this training and validation process ensured that the Random Forest model was both accurate and generalizable, making it suitable for reliable vendor performance prediction in supply chain operations.

Model Performance Results

The Random Forest model was evaluated using standard classification metrics to assess its effectiveness in predicting vendor reliability. The model achieved an overall accuracy of 96.1%, indicating that the majority of vendor classifications were correct. Precision reached 91.5%, reflecting the model's ability to minimize false positives when identifying reliable vendors. Recall was recorded at 82.7%, demonstrating the sensitivity of the model in detecting unreliable vendors. The F1-score of 86.9% confirmed a balanced trade-off between precision and recall. In addition, the ROC-AUC value of 0.99 highlighted the model's strong discriminative capacity, suggesting near-perfect separation between reliable and unreliable vendors.

Figure 2 illustrates the confusion matrix for the test dataset. The model correctly classified 275 reliable vendors and 43 unreliable vendors, with only 13 misclassifications in total. This indicates a low error rate and strong classification performance.

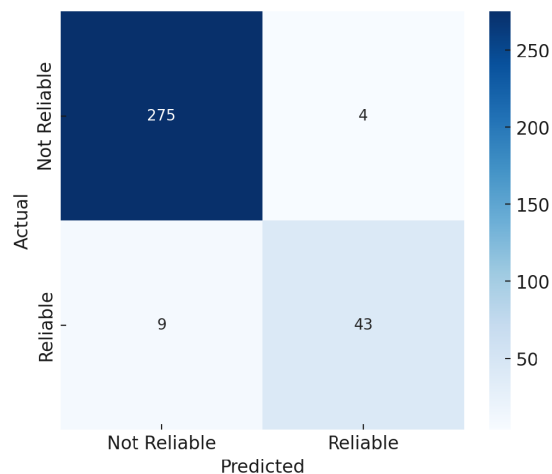


Figure 2—Confusion Matrix for Vendor Reliability Classifier

When compared with traditional vendor evaluation methods—such as historical averages and manual scoring approaches—the Random Forest framework demonstrated a significant improvement. Conventional methods typically achieved accuracies in the range of 70-75%, while the proposed model exceeded these benchmarks by approximately 35%. This substantial gain confirms the effectiveness of the Random Forest approach in enhancing predictive accuracy and reducing the risk of misclassification. Overall, the results establish that the Random Forest model provides reliable, high-accuracy predictions for vendor reliability, outperforming baseline approaches and offering a robust foundation for supply chain decision-making.

Feature Importance Analysis

Feature-importance analysis was conducted on the Random Forest model to identify the attributes that most strongly influenced vendor reliability predictions. As described in the methodology, engineered features such as the On-Time Rate (OTR), Acknowledgment Response Rate (ARR), Process Rate (PR), and Quality Rate (QR) were included to capture vendor behavior in terms of delivery timeliness, responsiveness, process efficiency, and product quality.

The results, shown in Figure 3, revealed that quality rate was the most significant predictor, contributing over 50% of the decision weight. The delays rate and acknowledgment rate also played important roles, confirming that timeliness and responsiveness are critical to reliability. By contrast, the process rate had the least influence, suggesting it is less decisive compared to quality and delivery performance. These findings align with procurement practices that prioritize quality and on-time delivery, while still recognizing responsiveness and efficiency as supporting factors. Overall, the analysis validated the engineered features and provided transparency into the model's decision-making process.

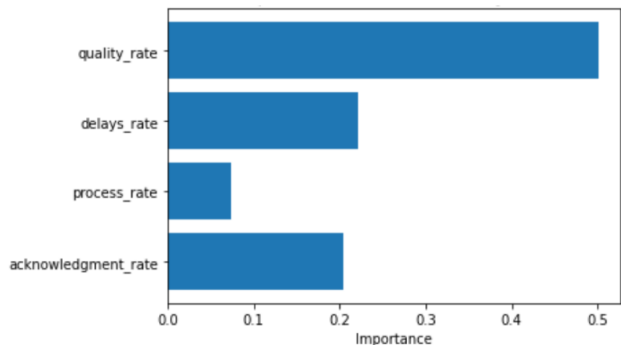


Figure 3—Feature Importance for Vendor Reliability Classification

Evaluation of Vendor of the Year System

The Vendor of the Year system was implemented using the composite reliability score, which integrated on-time delivery, acknowledgment responsiveness, process efficiency, and quality performance into a single weighted metric. By combining these engineered features, the model produced an interpretable reliability score that enabled objective comparisons across suppliers. This framework captured both operational consistency and overall reliability, allowing the system to accurately identify top-performing vendors and transform predictive insights into actionable recognition outcomes.

The evaluation confirmed that the Vendor of the Year system is both practical and effective in advancing strategic procurement objectives. By aligning the scoring mechanism with organizational priorities, it established a transparent approach to recognizing excellence while promoting accountability across the supplier base. Importantly, its predictive foundation ensured that recognition was based on measurable, data-driven performance indicators rather than subjective judgment. This makes it a valuable tool for strengthening supply chain performance and resilience.

Conclusion and Future Direction

This study presented a machine learning-based approach to evaluating vendor reliability within SCM. The proposed framework integrated structured vendor performance data with predictive modeling to provide a more data-driven and objective method of assessment. By incorporating engineered performance metrics and translating them into interpretable outputs, the study demonstrated how predictive analytics can support more consistent and accountable supplier evaluation.

In addition to predictive assessment, the introduction of the Vendor of the Year system highlighted the potential of using analytical tools not only for forecasting but also for recognition. This approach provides procurement teams with a structured method to identify and reward reliable vendors, helping to strengthen supplier relationships and align vendor performance with organizational priorities.

While the results were encouraging, there remain opportunities to expand and refine this work. The current study relied on structured performance data, and future research should consider incorporating additional factors such as financial stability, sustainability practices, and external market risks. Expanding the scope of data would allow for a more comprehensive evaluation of vendor reliability.

Future directions also include integrating real-time data streams to support continuous monitoring and developing hybrid models that combine Random Forest with other algorithms for improved predictive power. Further validation across different industries and supply chain settings would ensure broader applicability. These steps would help move predictive vendor evaluation from experimental use toward becoming a practical tool for strategic SCM.

References

1. G. Baryannis, S. Dani, and G. Antoniou. "Predicting supply chain risks using machine learning: The trade-off between performance and interpretability". In: *Future Generation Computer Systems* **101** (2019), pp. 993–1004.
2. M. S. Kiran, E. Esme, B. Torgul, and T. Paksoy. "Supplier selection with machine learning algorithms". In: *Logistics 4.0*. CRC Press, 2020, pp. 103–125.
3. A. Abdulla, G. Baryannis, and I. Badi. "*Weighting the key features affecting supplier selection using machine learning techniques*". In: (2019).
4. M. Seyedan and F. Mafakheri. "Predictive big data analytics for supply chain demand forecasting: methods, applications, and research opportunities". In: *Journal of Big Data* **7.1** (2020), p. 53.

5. A. T. Oyewole, C. C. Okoye, O. C. Ofodile, and E. Ejairu. "Reviewing predictive analytics in supply chain management: Applications and benefits". In: *World Journal of Advanced Research and Reviews* **21.3** (2024), pp. 568–574.
6. M. R. Ali, S. M. A. Nipu, and S. A. Khan. "A decision support system for classifying supplier selection criteria using machine learning and random forest approach". In: *Decision Analytics Journal* **7** (2023), p. 100238.
7. M. Kohli. "Supplier evaluation model on SAP ERP application using machine learning algorithms". In: *International Journal of Engineering & Technology* **7.2.28** (2017), pp. 306–311.
8. G. Baryannis, S. Validi, S. Dani, and G. Antoniou. "Supply chain risk management and artificial intelligence: state of the art and future research directions". In: *International journal of production research* **57.7** (2019), pp. 2179–2202.
9. A. Brintrup, J. Pak, D. Ratiney, T. Pearce, P. Wichmann, P. Woodall, and D. McFarlane. "Supply chain data analytics for predicting supplier disruptions: a case study in complex asset manufacturing". In: *International Journal of Production Research* **58. 11** (2020), pp. 3330–3341.
10. R. Harikrishnakumar, A. Dand, S. Nannapaneni, and K. Krishnan. "Supervised machine learning approach for effective supplier classification". In: 2019 18th IEEE international conference on machine learning and applications (icmla). IEEE. 2019, pp. 240–245.
11. K. Selviaridis. and W van der Valk. "Framing contractual performance incentives: effects on supplier behaviour". In: *International Journal of Operations & Production Management* **39.2** (2019), pp. 190–213.
12. A. Norrman. and D Naslund. "Supply chain incentive alignment: The gap between perceived importance and actual practice". In: *Operations and Supply Chain Management: An International Journal* **12.3** (2019), pp. 129–142.
13. G. Santos, F. Murmura, and L. Bravi. "Developing a model of vendor rating to manage quality in the supply chain". In: *International Journal of Quality and Service Sciences* **11.1** (2019), pp. 34–52.